

Crossbow Softball Pitcher

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Introduction:

The Goal of this project was to fashion a device that could accurately launch a standard softball onto a target within a range of distance. Given that no chemical energy could be used, we had to create a device that utilized kinetic and potential energy to launch the ball. We drew inspiration from crossbows and ballista to make a mounted crossbow that launches an item from a 'mitt' or basket, rather than the usual arrow.

Certain design features required substantial thought and planning. The angle of launch and the launch velocity determine the ball's path once shot, thus our design had to be able to manipulate those aspects of the ball's trajectory. We specifically designed the stand so that our crossbow would be able to rotate around both the y-axis and the x-axis. We also designed the crossbow so that the maximum amount of power would be generated for the softball launch. Using steel cables and fiberglass bow limbs we were able to combine the stiffness of the steel and the flexibility of the fiberglass to create a system that would generate the optimal amount of power for the crossbow. These methods to achieve measurable velocity and angle underwent numerous revisions as the design process progressed.

The significance of this project is that it combines measurable results with function and form. The launch velocity and angle allow us to estimate the landing location of a ball using kinematic equations. It's a technical application of physical principles. Understanding these principles allowed us to optimize our design for performance.

Methods and Materials:

Our method of design was inspired by crossbows and ballistae. After studying these mechanisms, we thoroughly understood how we could modify their design to suit softball-launching. We had to widen the firing barrel to suit the change in projectile and add a softball 'mitt' to hold the ball and prevent it from rolling through the firing process. This mitt, attached to a shuttle, glides on two metal rods that are connected to the bow limbs and the rear of the bow barrel. The metal rods the shuttle glides on are a design choice we made early on as a way to reduce the friction of the system. Another design choice we held to throughout the process was a 360-degree swiveling base and a range of launch angles between 0 and 60 degrees. This allowed for greater control over the trajectory of the ball and the final landing spot of the ball.

Some design choices that we made during construction and after our initial planning include the winch, the specific design of the articulation system around the x-axis, and the steel cable that attaches to the bow limbs. The articulation system was

designed after we began construction because we did not know what kind of load it would need to be able to handle. Thus, we could not design a system that was strong enough to support the bow at the different angles we would need to utilize. Once we knew those parameters we settled on a pin and axle design, which works by rotating the barrel of the gun on an axle, PVC pipe, with a pin inserted through the base posts to lock the crossbow in place. This allows for a range of angles to be achieved and greater control of the softball trajectory. The second design choice that we made during construction was the use of a winch. We were aware of the use of a winch to load historical crossbows and ballistae but did not believe there would be a need for one on our crossbow. However, upon completion, we realized that drawing the bow was extremely difficult and the winch allowed us to use angular mechanics to make drawing the bow significantly easier. The final design choice we made during construction was the use of steel cable as our bowstring. There were two reasons for this switch of material: we needed to make the bowstring more durable, and more importantly, the bowstring needed to be stiffer in order to draw the bow limbs far enough.

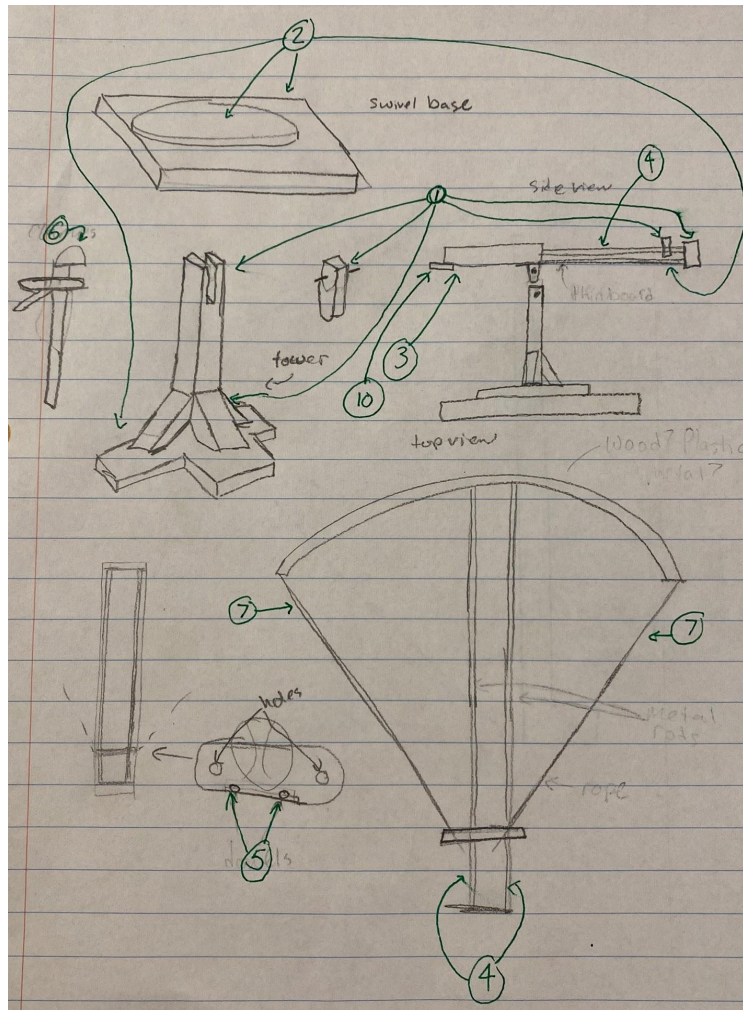


Image 1: Original sketch of the design plan. A lot stayed the same, although there were significant changes. The method of securing the bow to the tower changed, and the direction of the bow's arc also changed after we researched bows. We had to accommodate the softball by giving a gap between the arms of the bow.

After the original sketch (image 1) was completed, we set about modeling the design of the bow physically. We weren't sure if the bow arms would face forward or backward but knew they would pull backward no matter what. We made a model that highlighted features such as the tower pivot, shuttle attachment, and one feature we ditched—the height-adjusting tower. These features are shown in image 2.



Image 2: The functional prototype of our design, made out of popsicle sticks, twine, paper clips, and hot glue.



Image 5: Pin and pivot system that connects the tower and bow. Originally, many angles were chosen and drilled, but they turned out to be imprecise, so we found a couple of slots that were reliable and estimated their angles. The semicircle's connection to the bow was very important for precision, so we hot glued them together and then screwed them in place, guaranteeing a snug fit.

We had difficulty finding a string that would allow the bow to take on the force of pulling back. Pictured in image 6 are the pieces of the bow before assembly, including a white string we later replaced with a metal cord. Stringing the bow proved to be a challenge, even with the metal cord. Though the new string was better for distance and power, the force of the bow's draw slowly tore everything made of 2x4s and screws apart. This led to the crossbow's decommission before the end of the challenge.

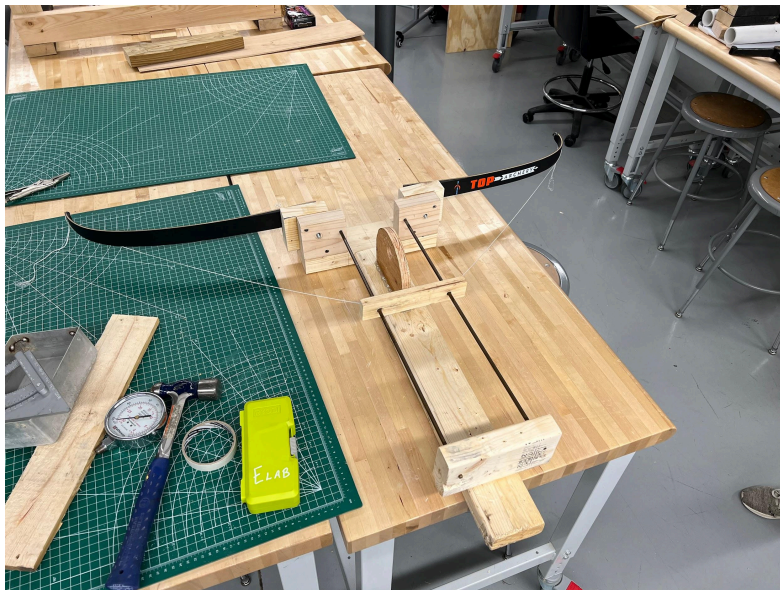


Image 6: The beginning of putting the systems together. We didn't know how much string we would need or how hard we would need to pull back on the string to shoot the ball adequately far. The string pictured did not put the bulk of the responsibility of stretching on the bow arms so we replaced it with a metal cord sheathed in plastic.

Figure 1: Bill of Materials

(the prices for the fasteners bought at Home Depot aren't included, but the bill was under \$4)

Name	Material	Measurements	Quantity	Source/Price
<i>Angle Measurement plate</i>	<i>1/8" plywood</i>	<i>2'x2'</i>	<i>1</i>	<i>Storage room</i>
<i>Base Plate</i>	<i>1" Plywood, cardboard circle</i>	<i>20" diameter</i>	<i>1</i>	<i>Storage room, recycled box</i>
<i>Tower half</i>	<i>2x4</i>	<i>18", with cuts</i>	<i>2</i>	<i>Storage Room</i>
<i>Tower supports</i>	<i>2x4</i>	<i>11", with cuts</i>	<i>4</i>	<i>Storage Room</i>
<i>Barrell</i>	<i>2x4</i>	<i>2 1/2"</i>	<i>1</i>	<i>Storage Room</i>
<i>Pin block</i>	<i>1" plywood</i>	<i>5"x6", with cuts</i>	<i>1</i>	<i>Storage Room</i>
<i>Tower Pin</i>	<i>1" diameter PVC pipe</i>	<i>5 1/2"</i>	<i>1</i>	<i>Storage Room</i>
<i>Tower Angle pin</i>	<i>4" Bolt</i>		<i>1</i>	<i>Home Depot</i>
<i>Riser Base</i>	<i>2x4</i>	<i>1'</i>	<i>1</i>	<i>Storage Room</i>
<i>Riser Block</i>	<i>2x4</i>	<i>3 3/4"</i>	<i>4</i>	<i>Storage Room</i>
<i>Bow fasteners</i>	<i>4" bolt, nuts and washers</i>		<i>2</i>	<i>Home depot</i>
<i>50 lb TopArchery Bow Limbs</i>	<i>wood/plastic</i>	<i>20 3/4" tip to ground unstretched</i>	<i>2</i>	<i>Found on Amazon, Approx \$60</i>
<i>Flight rod</i>	<i>1/8" diameter steel</i>	<i>2 1/2"</i>	<i>2</i>	<i>Physics supply closet</i>
<i>Shuttle</i>	<i>2x4</i>	<i>8"x1 1/2"x2"</i>	<i>1</i>	<i>Storage Room</i>
<i>Flight rod anchor</i>	<i>2x4</i>	<i>8 1/2"</i>	<i>1</i>	<i>Storage room</i>
<i>Mitt</i>	<i>Hot-glued popsicle sticks</i>	<i>Not measured, appropriate size for a softball</i>	<i>1</i>	<i>Supply cart</i>
<i>Winch</i>	<i>1" dowel, screw, pin</i>	<i>4"</i>	<i>1</i>	<i>Storage room</i>
<i>cut -rope</i>	<i>White polyester cord</i>	<i>2 1/2'</i>	<i>half a spool</i>	<i>Storage room</i>
<i>string</i>	<i>Metal cord</i>	<i>Approximately 5 1/2'</i>	<i>1</i>	<i>Storage room</i>

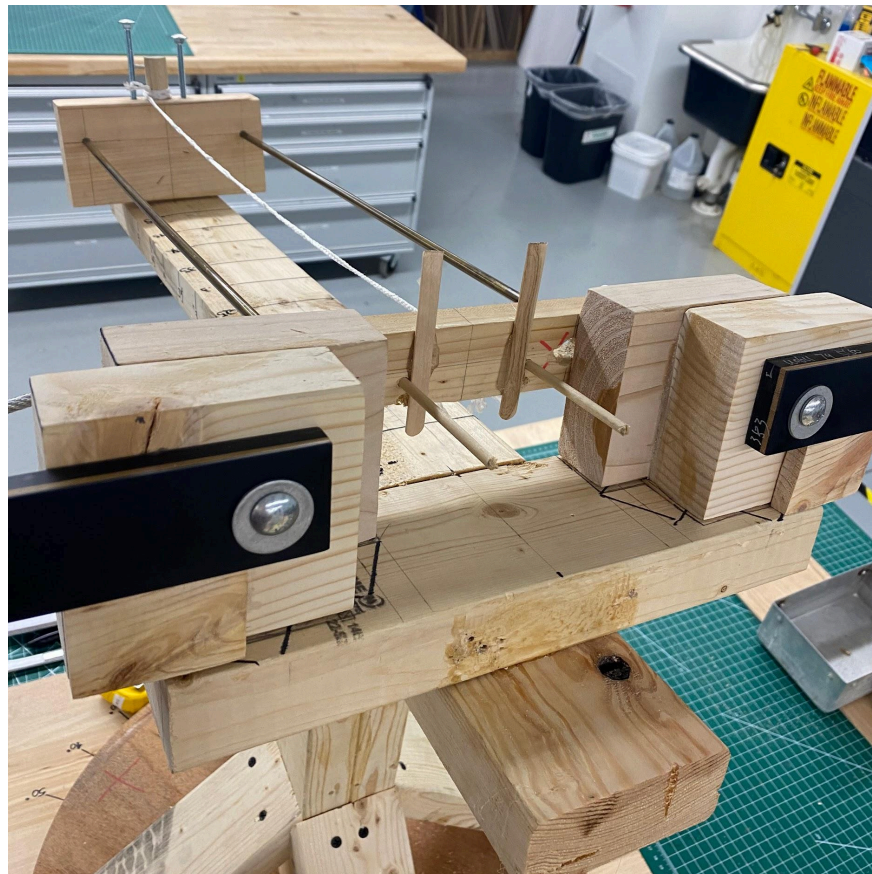


Image 7: The exit zone. This region proved to be the most difficult to design and build, as it undergoes a lot of stress and has specific requirements. The ball shouldn't roll down the track, or energy will be lost to friction and the motion of the ball will be more difficult to predict. We drilled the dowel rods in to act as a mitt and hold the ball in place. We also had to make sure the opening was big enough for the ball. Another challenge was that the blocks of wood needed to have enough space for the fasteners and rods. This led to the offset design. The arm segment was difficult to attach, as it was a high-stress area. We screwed both the barrel and the riser support to a piece of wood below with several screws to ensure structural integrity.



Image 7: The finished crossbow on competition day. The bow performed better than expected, though not through reliance on our analytical means. The MATLAB figure did not accurately predict our measured results, but it was fairly easy through trial and error to adjust the angle and launch speed to land the ball near the target.

Analyses and results:

We used energy methods to estimate the launch velocity of the ball given the angle of launch and the length the bow was pulled back, as seen in image 8.

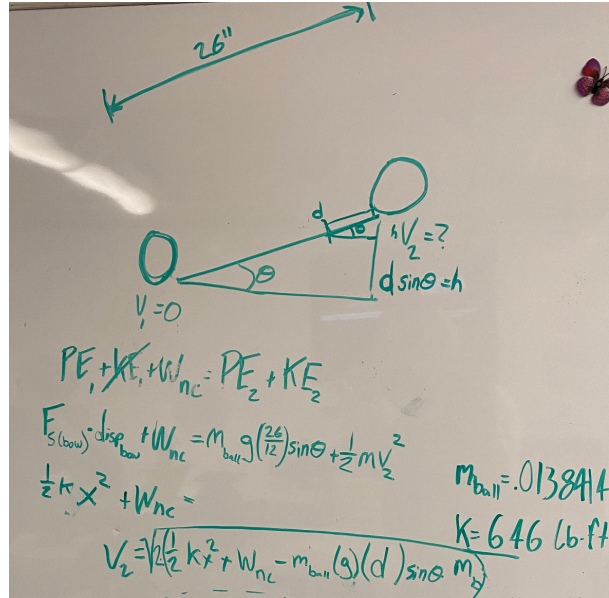


Image 8: energy methods calculating to find the exit velocity of the ball given the angle of launch, k constant of the bow, and draw distance. Many of the measurements above weren't precise—they are corrected in Matlab and reported in metric units.

We believe the relationship between length pulled back and distance traveled is accurate in its trend but not precise in its values. We also used approximation to estimate the spring force of the bow. We measured with a tape measure the displacement of the bow tips when the bow was fully drawn, assuming that it took 50 lb of force to get to that point, which gave us a k constant impacted by our imprecision in measuring. The equation we drew from the work in image 8 is

$$v_{exit} = \sqrt{2\left(\frac{1}{2}k(l)^2 + W_{nc} - m_{ball}^2(g)(l)\sin(\theta)\right)}$$

Where k is the spring constant of the bow, g is the acceleration due to gravity, l is the draw distance, and theta is the launch angle. Also important to note is that we assumed there was no nonconservative work because we had no way of finding it. We did our best to minimize nonconservative forces by placing heavy lead bricks on the base and dowel rods to minimize friction, rolling, and spin in the air. But our calculations were bound to be off because no nonconservative force could be completely disregarded. Our goal was to estimate the required length to draw back.

$$X(t) = \frac{V_0 \cos(\theta_0)}{c} (1 - e^{-ct})$$

$$V_0 = \sqrt{2 \left(-\frac{1}{2} kx^2 - m^2 g l \sin(\theta) \right)}$$

$$x(t) = \frac{V_0 \cos(\theta_0)}{c} (1 - e^{-ct})$$

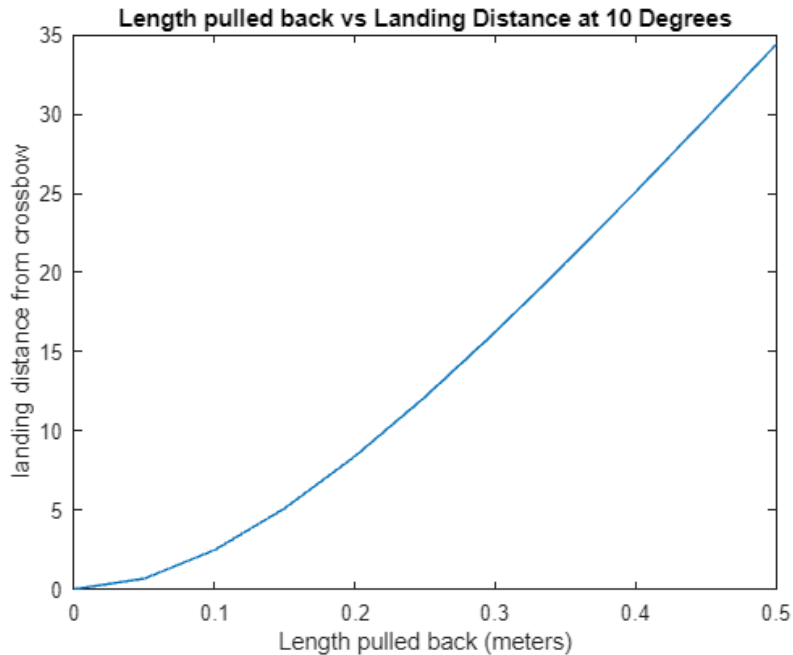
$$V_x = V_0 \cos(\theta_0) e^{-ct}$$

Image 9: Kinematic equations for X-Distance and Velocity. We used these formulas to predict the landing distance of the ball.

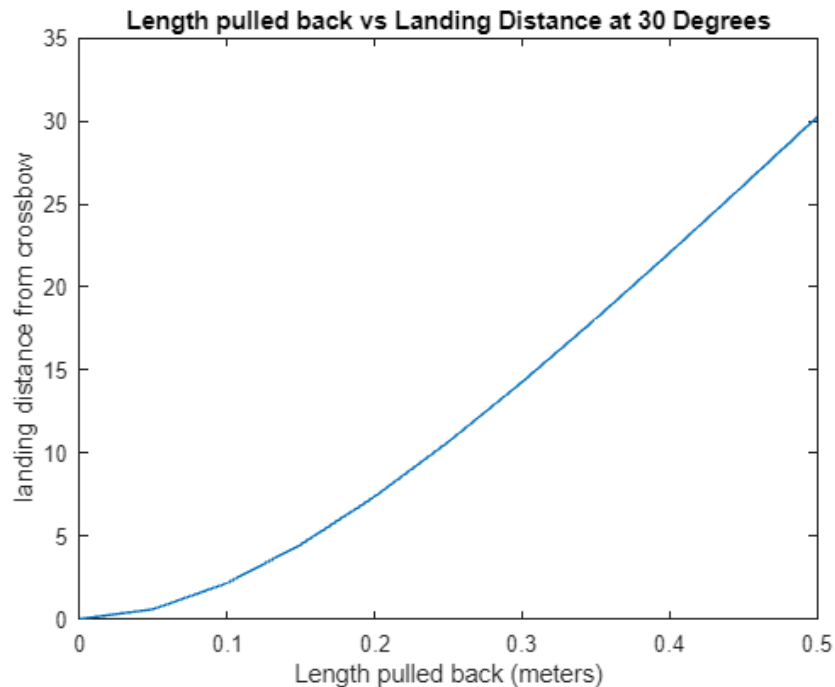
We used the kinematic equations (above in image 9) to find the displacement of the ball after launch. The key equation was $x(t) = \left(\frac{v_0 \cos(\theta)}{c} \right) (1 - e^{-ct})$

where v_0 was the launch velocity, θ was the angle of launch, c was the drag coefficient of a softball at a low speed, and t was time.

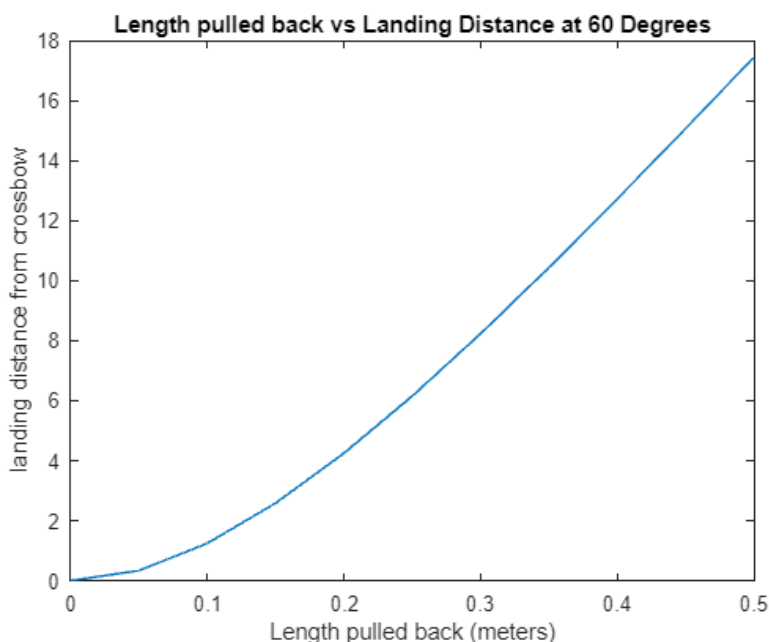
A significant amount of time was spent in an attempt to find $x(l)$, a function of displacement in terms of the draw distance. We finally gave up and tried plotting x versus l to see if the relationship looked plausible, with t and l as vectors of differing increments and start/end values but the same number of values. The plots of landing displacement given draw distance are plotted below in graphs 1-3 for each option of launch angle. Experimental calibration used the estimated landing distances as a starting point for the draw distance. The crossbow seemed to need a little more pull than what Matlab predicted in order to achieve the desired path of flight. The dowel rods and lead bricks mentioned before were involved in the process of optimization, and so were the change of the string used, adjustments in fastening components to each other, and the rearranging of the shuttle hooks to accommodate a thicker string. Each change set off a chain reaction of other changes to adapt the machine as a whole to the new part. As a result, the bow became a stronger and better version of the original and eventually became so powerful that 2x4s and drywall screws couldn't withstand the work done by repeated draws



Graph 1: MATLAB prediction for the bow when it was at a 10-degree angle. The units for both axes are meters. We obtained this graph and the rest through the use of kinematic and energy methods of analysis. The shape of the graph makes sense in hindsight because the relationship between the length pulled back and the force applied by the bow should look parabolic, and the force of the bow determines the exit velocity, so the landing distance would have a parabolic relationship with the length pulled back if there was no drag or loss of energy. The drag makes the ball land closer to the launch point.



Graph 2: MATLAB prediction for the bow when it was at a 30-degree angle. We knew from comparing this graph to the first that something was wrong with our code. Intuition tells us that a 30-degree incline will shoot the ball further than 10 degrees



Graph 3: MATLAB prediction for the bow when it was at a 60-degree angle. The 60-degree angle graph predicts that a ball pulled back from the same distance at 60 degrees will land closer than at 30 degrees.

Conclusions/Discussion:

The goal of this project was divided into two separate areas. The first was to design, test, and build a machine that could launch softballs at a target of varying distances. We achieved this through some medieval inspiration, making our own version of a ballista, modified to shoot softballs. Our design process was fairly straightforward, with no major design changes happening over the course of construction and testing. One of the design elements we tried to focus on was flexibility. We wanted our design to be able to adjust to any situation we might need. We did this through a method for adjusting the angle, as well as a mechanism for varying the draw distance of the “bow”. The biggest thing that allowed us to be flexible was the inclusion of a winch and holding mechanism. This allowed us to draw the “bow” to whatever distance we wanted, lock it in place, and make any adjustments needed to get a good shot. The second objective of this project was to come up with a Matlab simulation that could predict the trajectory of the shot based on some input variable. We decided that we would make our equations functions of draw distance and launch angle.

In real-world testing, our Matlab code did not accurately predict the distance the ball would travel. This was due to a variety of reasons. First, the equations that we used were correct for the trajectory of a ball with air resistance, however, the assumptions we made in order to allow our functions to be based on draw distance and angle were most likely wrong, or

at least deeply flawed. The assumptions we made were not random, however, and given more time, testing, and question-asking, we could come up with a more accurate simulation. One big assumption that was made that is most likely the main source of error was in the way we treated the “bow”. Because we used energy methods to calculate the initial velocity, we needed a way to treat the bow as a non-conservative force. We could not find an accurate way to measure the elasticity of the limbs, and the seller did not have any helpful information, so we treated it like a spring. In order to find a k constant, we used the rated draw weight, 50 lbs, and estimated the displacement of the tips of the bow under full draw. This allowed us to calculate a constant, however rough it may have been. The simplest way to improve the accuracy of our simulation would be to get a more accurate equation for the force applied by the bow on the shuttle. This would give a more accurate initial velocity, which would give more accurate distance predictions. Within the design itself, there are not that many major sources of error. The biggest factor in the distance would most likely be the shuttle design. We had trouble keeping it level underdraw, so during launch it may not have shot the ball perfectly straight. Instead, it may have added some unwanted vertical velocity that is not accounted for in our simulation. The other sources of error are the usual suspects, friction along moving parts, and vibrations due to the forces exerted during launch.

Overall, our design worked surprisingly well. During the competition we had difficulty hitting the target, but that was not due to flaws in the design, but rather our inability to properly calibrate our shots in such a limited time and shot count. If we were to do this project again, I don't think we would do a redesign, but rather refine our points of failure. During the building and testing phase, there were several issues that we ran into. The first, which we were able to fix, was that the connection between the “barrel” section and the “bow” section was not strong enough for the forces experienced during a full draw. We fixed this by adding further support beneath the “barrel” and “bow”, and this helped to increase reliability. The second point of failure, which ended up cutting our competition a few shots short, was the shuttle that held and launched the softball. During testing, there were two main points of failure on the shuttle. The first was the eye hooks that attached the string to the shuttle. In both testing and in the competition, after a few shots, the hooks were ripped out of the shuttle. There are two ways of fixing this issue. The first is we make the shuttle bigger so that there is more material surrounding the eye hooks. The second is to get eye hooks that are self-tapping. The ones used had short threads and needed a pilot hole and this certainly decreased their effectiveness. The second major point of failure on the shuttle was the mechanism used to hold the softball in place. During both testing and competition, at one point or another, the supports we had to hold the ball failed due to stress. We somewhat fixed this during the competition, but not in a way that was fitting to the rest of the design. One way to fix the problem would be to make the ball supports out of metal and fasten them to the shuttle with glue or some other fastener. Another issue that we ran into was inconvenient but not hugely problematic. Our angling mechanism was great in theory, but in practice, was not really useful. This is the part that would need the most improvement. We would either need to find a way to make the angle measurement more precise or come up with an entirely different mechanism for setting an angle.

More time would be devoted to accurately modeling the bow on Matlab in the future, as well. Our measurements, though as sound as we could make them, were imprecise. Our Matlab code was underdeveloped. Though it gave a good representation of the relationship between

Matlab will be attached to Schoology with this report. There was some difficulty attaching it to the document directly.



